

石嘴山市华岳新材料科技有限公司

SHIZUISHAN HUAYUE NEW MATERIAL TECHNOLOGY CO., LTD.

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安全技术说明书 • SAFETY DATA SHEET (MSDS/SDS)

氰氨化钙（石灰氮）• CALCIUM CYANAMIDE | 依据 GB/T 16483、GB/T 17519 及 GHS 编制 • Prepared per GB/T 16483, GB/T 17519 and GHS
| 文件编号 Doc. No. HY-SDS-CCN-001

第 1 部分：化学品及企业标识 Section 1 — Identification

化学品中文名 Chinese name	氰氨化钙（石灰氮）	化学品英文名 English name	Calcium cyanamide
CAS 号 / EC 号	156-62-7 / 205-861-8	分子式 / 分子量 Formula / MW	CaCN ₂ / 80.10
建议用途 Recommended use	化工中间体、冶金脱硫剂、缓释氮肥及土壤处理剂 Chemical intermediate, metallurgical desulphurising agent, slow-release nitrogen fertiliser and soil treatment agent		
限制用途 Restrictions	不得用于食品、饲料、药品用途 Not for food, feed or pharmaceutical use		
供应商 Supplier	石嘴山市华岳新材料科技有限公司 / Shizuishan Huayue New Material Technology Co., Ltd. 宁夏石嘴山市惠农区河滨工业园区兴园路东		
应急电话 Emergency telephone	公司值班电话 Company duty line: +86 952 3986888 中国国家化学事故应急咨询专线 China NRCC: 0532-83889090 消防/急救 Fire / Medical: 119 / 120		

第 2 部分：危险性概述 Section 2 — Hazards Identification

GHS 分类 GHS classification	遇水放出易燃气体的物质 类别 2；急性毒性—经口 类别 4；皮肤腐蚀/刺激 类别 2；严重眼损伤/眼刺激 类别 1；皮肤致敏物 类别 1；特异性靶器官毒性—一次接触 类别 3（呼吸道刺激）；危害水生环境—长期危害 类别 3 Substances which in contact with water emit flammable gases, Cat. 2; Acute toxicity (oral), Cat. 4; Skin corrosion/irritation, Cat. 2; Serious eye damage, Cat. 1; Skin sensitisation, Cat. 1; STOT-SE Cat. 3 (respiratory tract irritation); Hazardous to the aquatic environment — chronic, Cat. 3
象形图 Pictograms	GHS02（火焰）、GHS05（腐蚀）、GHS07（感叹号） GHS02 flame, GHS05 corrosion, GHS07 exclamation mark
警示词 Signal word	危险 / DANGER
危险性说明 Hazard statements	H261 遇水放出易燃气体 In contact with water releases flammable gases H302 吞咽有害 Harmful if swallowed H315 造成皮肤刺激 Causes skin irritation H317 可能引起皮肤过敏反应 May cause an allergic skin reaction H318 造成严重眼损伤 Causes serious eye damage H335 可能引起呼吸道刺激 May cause respiratory irritation H412 对水生生物有害并具有长期持续影响 Harmful to aquatic life with long lasting effects
防范说明	P223 + P231 + P232 避免接触水；在惰性气体下操作；防潮

Precautionary statements	Do not allow contact with water; handle under inert gas; protect from moisture P260 / P264 不要吸入粉尘；作业后彻底清洗 Do not breathe dust; wash thoroughly after handling P270 / P271 使用时不得进食、饮水或吸烟；只能在室外或通风良好处使用 Do not eat, drink or smoke when using; use only outdoors or in a well-ventilated area P280 戴防护手套、防护服、护目镜及面罩 Wear protective gloves, clothing, eye and face protection P301+P312 / P305+P351+P338+P310 食入：不适时立即就医；眼睛：用水小心冲洗数分钟，取出隐形眼镜后继续冲洗，立即呼叫解毒中心或就医 IF SWALLOWED: call a POISON CENTER if you feel unwell; IF IN EYES: rinse cautiously with water for several minutes, remove contact lenses, continue rinsing, immediately call a POISON CENTER or doctor P302+P334 / P333+P313 皮肤沾染：刷掉颗粒后浸入冷水；出现皮肤刺激或皮疹时就医 IF ON SKIN: brush off loose particles, immerse in cool water; if skin irritation or rash occurs, get medical advice P402+P404 / P501 存放在干燥处，密闭容器中；按国家和地方法规处置内容物和容器 Store in a dry place in a closed container; dispose of contents and container per national and local regulations
其他危害 Other hazards	接触本品期间饮酒可引起面部潮红、心悸等类双硫仑（戒酒硫）反应；粉尘在受限空间可能形成爆炸性混合物 Alcohol consumption in connection with exposure may cause an antabuse-like reaction (facial flushing, palpitations); dust may form explosive mixtures in confined spaces

第 3 部分：成分/组成信息 Section 3 — Composition / Information on Ingredients

组分 Component	CAS 号 CAS No.	含量 Content	说明 Remarks
氰氨化钙 Calcium cyanamide	156-62-7	≥ 55 %（以有效氮 ≥21.5 % 计）	主成分 Main constituent
氧化钙 / 氢氧化钙 Calcium oxide / hydroxide	1305-78-8 / 1305-62-0	余量 Balance	工艺伴生 Process-related
碳（石墨）Carbon (graphite)	7782-42-5	< 15 %	工艺伴生 Process-related
碳化钙 Calcium carbide	75-20-7	≤ 0.2 %	受控杂质 Controlled impurity

组分含量为典型范围，随批次略有变化。Component contents are typical ranges and vary slightly between batches.

第 4 部分：急救措施 Section 4 — First-aid Measures

吸入 Inhalation	迅速脱离现场至空气新鲜处，保持呼吸道通畅；呼吸困难给氧，呼吸停止立即人工呼吸并就医 Move to fresh air, keep airway clear; give oxygen if breathing is difficult, give artificial respiration if breathing has stopped, seek medical attention
皮肤接触 Skin contact	立即刷去干燥粉末（勿先用水冲），脱去污染衣物，再用大量流动清水冲洗至少 15 分钟；就医 Brush off dry powder first (do not wet immediately), remove contaminated clothing, then flush with plenty of running water for at least 15 minutes; seek medical attention
眼睛接触 Eye contact	提起眼睑，用大量流动清水或生理盐水冲洗至少 15 分钟，立即就医 Hold eyelids open and rinse with plenty of running water or saline for at least 15 minutes; obtain immediate medical attention
食入 Ingestion	漱口，禁止催吐；给患者饮少量水；意识不清者不得经口给予任何物质；立即就医 Rinse mouth; do NOT induce vomiting; give small amounts of water; never give anything by mouth to an unconscious person; obtain medical attention immediately
对医生的提示 Note to physician	对症与支持治疗；注意类双硫仑反应（面部潮红、低血压、心动过速），患者接触后 24 小时内应避免饮酒

	Treat symptomatically and supportively; watch for antabuse-like reaction (flushing, hypotension, tachycardia); patient should avoid alcohol for 24 h after exposure
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第 5 部分：消防措施 Section 5 — Fire-fighting Measures

合适灭火介质 Suitable media	干砂、干粉（D 类）、二氧化碳 Dry sand, Class D dry powder, carbon dioxide
禁用灭火介质 Unsuitable media	严禁使用水、水成膜泡沫及含水灭火剂 Never use water, water jets, AFFF or any water-based extinguishing agent
特别危险性 Special hazards	遇水生成乙炔、氨等易燃气体，可能引发爆炸；受热分解释放一氧化碳、二氧化碳、氮氧化物及氰化物 Contact with water generates flammable acetylene and ammonia which may explode; thermal decomposition releases CO, CO ₂ , nitrogen oxides and cyanides
消防人员防护 Protection for firefighters	佩戴正压式空气呼吸器及全身防护服；上风向灭火；尽可能将容器移离火场，用干砂覆盖隔离 Wear positive-pressure SCBA and full protective clothing; fight fire from upwind; move containers from the fire area if safe, blanket with dry sand

第 6 部分：泄漏应急处理 Section 6 — Accidental Release Measures

人员防护 Personal precautions	隔离泄漏区并限制出入，应急人员佩戴防尘面具、护目镜及防护服；避免扬尘 Isolate and restrict access; responders to wear dust respirator, goggles and protective clothing; avoid dust generation
环境防护 Environmental precautions	防止流入下水道、地表水及地下水 Prevent entry into drains, surface water and groundwater
清理方法 Methods for containment	用清洁工具干法收集（禁止用水冲洗）至干燥、有标识的密闭容器，交有资质单位处置；残余物用干砂吸收后清除 Collect dry with clean tools (never flush with water) into a dry, labelled, closed container for disposal by a licensed contractor; absorb residues with dry sand and remove

第 7 部分：操作处置与储存 Section 7 — Handling and Storage

操作处置 Handling	密闭操作，局部排风；避免产生粉尘；作业现场严禁烟火、禁止饮食饮酒；防止与水、酸、氧化剂接触；空容器仍残留有害物 Handle in closed systems with local exhaust ventilation; avoid dust; no smoking, eating or alcohol at the workplace; prevent contact with water, acids and oxidisers; empty containers retain hazardous residues
储存 Storage	储存于阴凉、干燥、通风良好的专用库房，温度<30℃、相对湿度<60%；远离火种热源；与酸类、氧化剂、水源分储；地面高于室外并设防水措施；配备泄漏应急处理设备 Store in a cool, dry, well-ventilated dedicated warehouse below 30℃ and <60% RH; away from ignition and heat sources; segregate from acids, oxidisers and water; raised, waterproofed flooring; provide spill-response equipment

第 8 部分：接触控制/个体防护 Section 8 — Exposure Controls / Personal Protection

职业接触限值 Occupational exposure limits	中国 GBZ 2.1：未列入（按总粉尘 PC-TWA 8 mg/m ³ 管理）；ACGIH TLV-TWA 0.5 mg/m ³ ；欧盟工作者 DNEL（吸入，长期）1 mg/m ³ China GBZ 2.1: not listed (manage as total dust, PC-TWA 8 mg/m ³); ACGIH TLV-TWA 0.5 mg/m ³ ; EU worker DNEL (inhalation, long-term) 1 mg/m ³ ; dermal DNEL 0.65 mg/kg bw/day
工程控制 Engineering controls	密闭化生产、局部机械排风；设置洗眼器与应急淋浴 Enclosed process with local mechanical exhaust; eyewash station and safety shower nearby
呼吸系统防护 Respiratory protection	KN95/P2 及以上防尘口罩；高浓度或应急时佩戴正压式空气呼吸器 KN95/P2 or better dust respirator; positive-pressure SCBA for high concentrations or emergencies
眼面 / 手 / 身体防护 Eye / hand / body	化学安全护目镜或面罩；耐碱防护手套（丁腈或氯丁橡胶）；防尘工作服及工作鞋，工作服单独存放清洗

Chemical safety goggles or face shield; alkali-resistant nitrile or neoprene gloves; dust-proof coveralls and footwear, laundered separately

第 9 部分：理化特性 Section 9 — Physical and Chemical Properties

外观与性状 Appearance	灰黑色粉末或颗粒 Grey-black powder or granules	气味 Odour	轻微氨味 Slight ammoniacal odour
熔点 Melting point	约 1145–1217 °C	沸点 Boiling point	不适用（分解） Not applicable (decomposes)
pH 值 pH	约 4.8（1 % 水悬浮液）	自燃温度 Auto-ignition	> 850 °C
蒸气压 Vapour pressure	0.51 Pa（20 °C）	典型粒径 Typical particle size	约 40 µm；180 µm 筛余物 ≤10 %
溶解性 Solubility	与水反应；不溶于常见有机溶剂 Reacts with water; insoluble in common organic solvents		

第 10 部分：稳定性和反应性 Section 10 — Stability and Reactivity

稳定性 Stability	干燥密闭条件下稳定 Stable under dry, closed conditions
应避免的条件 Conditions to avoid	受潮、遇水、明火、高温、粉尘扬散 Moisture, water, open flame, high temperature, dust dispersion
禁配物 Incompatible materials	水、酸类、强氧化剂、醇类 Water, acids, strong oxidisers, alcohols
危险分解产物 Hazardous decomposition	乙炔、氨、单氰胺、一氧化碳、氮氧化物、氰化物 Acetylene, ammonia, cyanamide, carbon monoxide, nitrogen oxides, cyanides

第 11 部分：毒理学信息 Section 11 — Toxicological Information

急性经口毒性 Acute oral toxicity	LD ₅₀ （大鼠）约 765 mg/kg（急性毒性估计值 ATE） LD ₅₀ (rat) approx. 765 mg/kg (ATE)
急性吸入毒性 Acute inhalation	ATE > 0.155 mg/L（4 h）
皮肤 / 眼刺激 Skin / eye	造成皮肤刺激；造成严重眼损伤（不可逆） Causes skin irritation; causes serious, irreversible eye damage
致敏性 Sensitisation	皮肤致敏物（类别 1） Skin sensitiser (Cat. 1)
特异性靶器官毒性 STOT	一次接触：呼吸道刺激（类别 3） Single exposure: respiratory tract irritation (Cat. 3)
致癌 / 致突变 / 生殖毒性 CMR	现有资料未将本品分类为致癌、致突变或生殖毒性物质 Not classified as carcinogenic, mutagenic or toxic to reproduction on available data
其他 Other	抑制乙醛脱氢酶，接触后饮酒可产生类双硫仑反应 Inhibits aldehyde dehydrogenase; alcohol after exposure can produce an antabuse-like reaction

第 12 部分：生态学信息 Section 12 — Ecological Information

水生毒性 Aquatic toxicity	鱼类慢性 LC ₅₀ 11.8 mg/L（21 d）；无脊椎动物慢性 EC ₅₀ 0.41 mg/L（21 d） Fish chronic LC ₅₀ 11.8 mg/L (21 d); invertebrate chronic EC ₅₀ 0.41 mg/L (21 d)
预测无效应浓度 PNEC	淡水 10 µg/L；海水 1 µg/L

降解与蓄积 Degradation / bioaccumulation	在环境中水解为单氰胺、尿素并最终矿化为氮化合物；无显著生物蓄积性 Hydrolyses in the environment to cyanamide and urea and ultimately mineralises to nitrogen compounds; no significant bioaccumulation potential
PBT / vPvB 评估	不属于 PBT 或 vPvB 物质 Not a PBT or vPvB substance

第 13 部分：废弃处置 Section 13 — Disposal Considerations

废弃物处置 Waste disposal	按危险废物管理，交由有资质单位处置；可在受控条件下缓慢水解中和后按当地法规排放；严禁随意倾倒或排入水体 Manage as hazardous waste and dispose of through a licensed contractor; may be slowly hydrolysed and neutralised under controlled conditions and discharged per local regulations; never dump or discharge into watercourses
污染包装 Contaminated packaging	空包装残留物按本品处理；清洁后方可回收 Residues in empty packaging must be treated as product; recycle only after cleaning

第 14 部分：运输信息 Section 14 — Transport Information

项目 Item	联合国危规 / ADR / RID / IMDG / IATA
联合国编号 UN number	UN 1403
联合国运输名称 UN proper shipping name	氰氨化钙（含碳化钙>0.1 %） CALCIUM CYANAMIDE with more than 0.1 % calcium carbide
危险性类别 Transport hazard class	4.3 — 遇水放出易燃气体的物质 / Substances which in contact with water emit flammable gases
包装类别 Packing group	III
海洋污染物 Marine pollutant	是 / Yes
特殊防范 Special precautions	严格防水防潮；禁与酸类、氧化剂混装；运输车辆须配备干砂等应急器材 Protect strictly from water and moisture; do not load with acids or oxidisers; vehicles to carry dry sand and spill equipment

若碳化钙含量 ≤0.1 %，运输分类可能不同，请以实际批次检验数据及承运人确认为准。If the calcium carbide content is ≤0.1 %, the transport classification may differ; confirm against the actual batch data and with the carrier.

第 15 部分：法规信息 Section 15 — Regulatory Information

中国 China	《危险化学品安全管理条例》；《危险化学品目录》（本品列入）；GB 6944 危险货物分类；GB 30000 系列化学品分类和标签规范；GB 190 危险货物包装标志 Regulations on the Safe Management of Hazardous Chemicals; Catalogue of Hazardous Chemicals (listed); GB 6944; GB 30000 series; GB 190
产品标准 Product standard	HG/T 5922-2021 《工业氰氨化钙》
境外 Overseas	EU REACH 已注册（EC 205-861-8）；CLP 法规（EC）No 1272/2008 分类与标签；US TSCA 清单收录；请使用方自行确认目的国登记与进口要求 Registered under EU REACH (EC 205-861-8); classified and labelled under CLP (EC) No 1272/2008; listed on the US TSCA Inventory. Users must verify registration and import requirements in the destination country.

第 16 部分：其他信息 Section 16 — Other Information

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编制依据 Basis of preparation	GB/T 16483-2008、GB/T 17519-2013、GB 30000 系列、GHS（第 10 修订版）及公开毒理与生态毒理资料

	GB/T 16483-2008, GB/T 17519-2013, GB 30000 series, GHS Rev.10, and publicly available toxicological and ecotoxicological data
缩略语 Abbreviations	ATE 急性毒性估计值；DNEL 推导无作用水平；PNEC 预测无效应浓度；SCBA 正压式空气呼吸器；STOT-SE 特异性靶器官毒性—一次接触；PBT 持久性、生物蓄积性和毒性物质

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